ONE Clinical Lymphedema Protocol

POSSIBLE F-TAG/QMS

FEDERAL TAGS (F-TAGS)

- **F 697:** Pain management
- **F 676:** Maintain ADL, Mobility, and Communication Abilities
- **F 684:** Maintain Physical, Mental, and Psychosocial Well Being
- **F 686:** Treatment/services to prevent pressure/heal PU's
- **F 684:** Quality of Care

Quality Measures

- Moderate/Severe Pain
- Help with ADL
- Walking/Locomotion
- ER Visits Hospitalizations
- Short & Long Stay Pressure Injuries

THERAPIES

- **PT**
- **OT**
- **RN**
- **LPN**

PURPOSE OF THERAPY

- To prevent wound development and preserve highest level of skin integrity
- To expediate or optimize edema control
- To facilitate a **ONE Clinical** lymphedema approach at facilities for best outcome
- To promote patient and caregiver education and understanding of disease process to be able to self-manage and reduce risk of infection and hospitalization
- To reduce the risk of complications associated with disease processes (infections, wounds, pain, etc.)

GOAL EXAMPLES

PT/OT

- Patient will exhibit decreased edema in (list body part) as measured by circumferential/volumetric reading of uninvolved to involved extremity and improved function in the areas of...
- Patient will express an understanding of the disease process as it relates to function and will demonstrate 100% return demonstration in home program techniques.
- Patient will increase AROM to/by (degrees in extremity) to increase patient performance of UE/LE dressing/bathing to supervision level.
- Patient will report decreased pain on (list standard pain scale) in R LE in order to improve participation in transfers to partial assistance.
- Patient will perform decongestive exercises (e.g., diaphragmatic breathing, pelvic tilts, ankle pumps) with supervision and minimal verbal cues to promote lymph flow.
- Patient will tolerate compression garment application for at least 6 hours daily without skin irritation.

TOOLS, ASSESSMENTS AND IMPLEMENTATION

Tools and Assessment

Limb circumferential measurements with volumetric calculations:

- Using a standard tape measure, circumferential measures are taken at pre-determined distances along the length of the limb
- Each circumference is used in a geometric formula for a cylinder which enables calculation of the limb volume

Pitting Edema Scale:

Rates tissue response to pressure:

- 0: No pitting (no indentation remains after pressure)
- 1+: Slight pitting (indentation up to 2 mm deep, rebounds immediately)
- 2+: Moderate pitting (indentation 3-4 mm deep, rebounds in 10-15 seconds)
- 3+: Deep pitting (indentation 5-6 mm deep, rebounds in 15-60 seconds)
- 4+: Very deep pitting (indentation 7-8 mm deep or more, rebounds in over 60 seconds)

Stemmer's Sign:

- A diagnostic test for lymphedema: inability to pinch the skin at the base of the second toe or finger

POSSIBLE F-TAG/QMS	THERAPIES	PURPOSE OF THERAPY	GOAL EXAMPLES	TOOLS & ASSESSMENTS
			• Patient will show a reduction in limb circumference by at least 2 cm through consistent use of to reduce the risk of UE/LE wounds. • Patient will demonstrate independence with ability to manage compression therapy, including donning/doffing garments and monitoring skin integrity. • Patient will perform self-care tasks (e.g., dressing, bathing) using adaptive equipment with minimal assistance. • Patient will demonstrate independence with ability to maintain skin integrity during ADLs by applying appropriate moisturizers and inspecting for signs of breakdown. • Patient will increase endurance to safely tolerate standing activity for three minutes for self-care and other life skill tasks. • Patient will demonstrate 100% understanding/compliance of scar reduction techniques as evidence by flat, scar tissue and improved/no loss of ROM during _____ (activity or time).	**Ankle Brachial Index (ABI)** • > 1.30 - Noncompressible. Calcified, likely from long-standing diabetes • 1.00-1.29 - Normal • 0.91-0.99 - Borderline. Consider risk and symptoms before compression • 0.81-0.90 - Mild PAD, possible intermittent claudication. Compression appropriate if asymptomatic • 0.5-0.80 - Moderate PAD, claudication expected. Compression if validated by vascular referral • < 0.5 - Significant perfusion change. Compression is not recommended. High risk and vascular assessment needed **Lymphedema Staging** Based on the International Society of Lymphology: • Stage 0: Latent-subclinical, no visible swelling • Stage 1: Spontaneously reversible swelling with elevation • Stage 2: Spontaneously irreversible swelling that does not resolve with elevation, fibrosis begins • Stage 3: Lymphostatic Elephantiasis-severe swelling, non-pitting, skin changes (e.g., papillomatosis) **Golematis Staging:** Staging severity of Lymphedema • <2 centimeters - Mild lymphedema • <2.1 to 4 centimeters - Moderate lymphedema • > 4.1 centimeters - Severe lymphedema **Skin and Tissue Assessment** • Texture (soft vs. indurated/fibrotic) • Presence of scars, radiation fibrosis, or wounds

POSSIBLE F-TAG/QMS	THERAPIES	PURPOSE OF THERAPY	GOAL EXAMPLES	TOOLS & ASSESSMENTS

Bioimpedance Spectroscopy (BIS) & Perometry

- Performed outside of facility
- BIS measures extracellular fluid levels and is useful for early detection and monitoring
- Perometry uses infrared light to measure limb volume

Intervention/Implementation

Manual Therapy:

- Manual Lymphatic Drainage (MLD) which stimulates the movement of fluids in the tissues using gentle massage
- Comprehensive Decongestive Therapy (CDT) which is a combination of MLD, bandaging, skin care and exercises

Compression:

- Multi-layer compression wraps (keep in mind Rule of LaPlace)
- Velcro garments
- Stockings
- IPC -Intermittent pneumatic compression therapy (pump/garment)

Positioning:

- Limb elevation- positioning the affected arm or leg above the heart to help drain lymph fluid and reduce swelling

Exercise:

- Low intensity and repetitive-should be done with compression bandages on the limb
- Encourage gentle muscle pumping, which aids the absorption and transport of the lymphatic fluid out of the limb
- Ankle pumps, ROM, sit to stands, weighted/resistance band strengthening exercises, ambulation

POSSIBLE F-TAG/QMS	THERAPIES	PURPOSE OF THERAPY	GOAL EXAMPLES	TOOLS & ASSESSMENTS
				Modalities: E-stim: Contraindication: previously radiated - Neuromuscular Electrical Stimulation (NMES) - Functional Electrical Stimulation (FES) - Transcutaneous Electrical Nerve Stimulation (TENS) - Microcurrent Therapy **Skin Care:** - Promotes healthy tissue and prevents skin breakdown and potential infections - Moisturize the tissue, keep it clean, and attend to small cuts to prevent infections or chronic wounds **Self-care:** - Self-MLD (Manual Lymphatic Drainage) - Consistent Compression Garment and/or Bandaging - Regular Exercise - Meticulous Skin Care - Regular Follow-up Visits with Medical Provider and/or Therapist - Be aware of constrictive clothing and heat to involved limb to decrease risk of recurring lymphedema **National Lymphedema Network**